

lie groups 1

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September 2026

1 Basics

Claim: The only continuous group homomorphisms $f : S^1 \rightarrow S^1$, where $S^1 \leq \mathbb{C}^\times$ is given by $f(z) = z^n$ for some fixed $n \in \mathbb{Z}$.

Proof: Denote by $p : \mathbb{R} \rightarrow S^1$ the exponential map given by $p(t) = e^{2\pi it}$. Since $\pi_1(\mathbb{R}, 0) = 0$, we can lift $f \circ p$ to $\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}$ such that $\tilde{f}(0) = 0$ and $p \circ \tilde{f} = f \circ p$. We have

$$p(\tilde{f}(s) + \tilde{f}(t)) = p(\tilde{f}(s))p(\tilde{f}(t)) = f(p(s))f(p(t)) = f(p(s)p(t)) = f(p(s+t)) = p(\tilde{f}(s+t))$$

so $\tilde{f}(s+t) - \tilde{f}(s) - \tilde{f}(t) \in \mathbb{Z}$ for all s, t . Fix s and vary t . The LHS is a continuous function and $\mathbb{Z}$ is discrete, so the LHS is constant for variable t . In particular, $\tilde{f}(s+t) - \tilde{f}(s) - \tilde{f}(t) = 0$ (by evaluating at $t = 0$), so $\tilde{f}$ is a (continuous) homomorphism of $\mathbb{R}$. This is the Cauchy functional equation, so $\tilde{f}$ has the form $\tilde{f}(t) = \alpha t$ for some $\alpha \in \mathbb{R}$. Since $f(1) = 1$, we have

$$p(0) = 1 = f(1) = f(p(1)) = p(\tilde{f}(1))$$

so $\alpha = \tilde{f}(1) = n \in \mathbb{Z}$. Thus, $f(z) = f(p(t)) = p(\tilde{f}(t)) = p(nt) = p(t)^n = z^n$, as desired.

Claim: The map $f : \mathbb{R} \rightarrow S^1 \times S^1 = T^2$ by $f(t) = (e^{2\pi it}, e^{2\pi \alpha it})$ for $\alpha \notin \mathbb{Q}$ is an embedding of Lie groups.

Proof: We can check smoothness and that group structure is preserved. Suppose for $t \neq 0$ that $f(t) = (1, 1)$, so $t \in \mathbb{Z}$ and $\alpha t \in \mathbb{Z}$. Then $\alpha = (\alpha t)/t \in \mathbb{Q}$, absurd.

Claim: $SL(n, \mathbb{R})$ is a regular submanifold of $GL(n, \mathbb{R})$ of dimension $n^2 - 1$.

Proof: The map $\det : GL(n, \mathbb{R}) \rightarrow \mathbb{R}^\times$ is C^∞ . Write $\det A = \sum_{k \in [n]} (-1)^{i+k} a_{i,k} \det M_{i,k}$, where $M_{i,k}$ is the minor of A formed by deleting the i th row and k th column. Suppose that $A \in \det^{-1}(1)$. Thus, $\frac{\partial \det A}{\partial a_{i,j}} = (-1)^{i+j} \det M_{i,j}$. If this number is zero for all i, j , in particular this partial is zero for a row of i 's and j 's (ie for all j for some fixed i). Then $\det A = 0$, which is absurd. Thus, 1 is a regular value of $\det$, so $SL(n, \mathbb{R}) = \det^{-1}(1)$ is an $n^2 - 1$ -dimensional submanifold.

2 The Exponential Map

Claim: Let X be a topological space and $E \leq X$. TFAE:

- (i). $E = A \cap U$ for A closed and U open.
- (ii). For each $x \in E$, there exists $U(x)$ such that $E \cap U$ is closed in U .
- (iii). E is open in $\overline{E}$.
- (iv). $\overline{E} - E$ is closed in X .
- (v). E is the difference of two closed sets in X .
- (vi). E is the difference of two open sets in X .

(i) $\rightarrow$ (ii). Then $E = E \cap U$ is closed in U .

(ii) $\rightarrow$ (vi).

3 Vector Bundles

Claim: Let $\pi : E \rightarrow M$ and $\pi' : E' \rightarrow M$ be vector bundles. Suppose $\varphi : E \rightarrow E'$ is fiber-preserving, linear over fibers, C^∞ , and bijective. Then φ is a diffeomorphism.

Proof: Pick $U \leq M$ such that $\phi : E|_U \rightarrow U \times \mathbb{R}^r$ and $\phi' : E'|_U \rightarrow U \times \mathbb{R}^r$ are local trivializations of π, π' . It suffices to show that the composition

$$U \times \mathbb{R}^r \xrightarrow{\phi^{-1}} E|_U \xrightarrow{\varphi} E'|_U \xrightarrow{\phi'} U \times \mathbb{R}^r$$

is a diffeomorphism. Denote this map by ψ . Note that $\psi : U \times \mathbb{R}^r \rightarrow U \times \mathbb{R}^r$ fixes U and for fixed $p \in U$ acts as a linear transformation $L_p : \mathbb{R}^r \rightarrow \mathbb{R}^r$, by identifying $\mathbb{R}^r = p \times \mathbb{R}^r$. Since φ is bijective, so is L_p . Thus, ψ is a C^∞ bijective map, whose rank at $(p, v) \in U \times \mathbb{R}^r$ is $\dim U + \dim \mathbb{R}^r = \dim(U \times \mathbb{R}^r)$, so ψ is full-rank. Thus, ψ is a diffeomorphism.

Slight mistake: $U, \mathbb{R}^r$ aren't independent. We can't say $\dim U + \dim \mathbb{R}^r = \dim(U \times \mathbb{R}^r)$, and then we're done. U is independent of $\mathbb{R}^r$, but $\mathbb{R}^r$ need not be independent of U . The Jacobian at (p, v) has the shape

$$D\psi_{(p,v)} = \begin{bmatrix} I & 0 \\ \partial_p L_p v & L_p \end{bmatrix}$$

which is invertible.

Claim: A manifold M has a global smooth frame $(X_1, X_2, \dots, X_n)$ iff M is parallelizable, ie $TM \cong M \times \mathbb{R}^n$ as vector bundles.

Proof: Suppose $(X_1, X_2, \dots, X_n)$ is a smooth frame. Define $\varphi : M \times \mathbb{R}^n \rightarrow TM$ by

$$\varphi(p, a^1, a^2, \dots, a^n) = \left(p, \sum_{i \in [n]} a^i X_i|_p \right)$$

We can check that φ is bijective, fiber-preserving, and linear over fibers. It remains to show that φ is C^∞ . First, note that the composition

$$M \times \mathbb{R}^n \xrightarrow{(X_1, X_2, \dots, X_n) \times 1_{\mathbb{R}^n}} TM^n \times \mathbb{R}^n \rightarrow (\mathbb{R} \times TM)^n \rightarrow TM^n$$

has image contained in $TM^{\oplus n}$ and is C^∞ since scalar multiplication $\mathbb{R} \times TM \rightarrow TM$ is C^∞ . Thus, we can restrict to $M \times \mathbb{R}^n \rightarrow TM^{\oplus n}$. Since addition $TM^{\oplus n} \rightarrow TM$ is C^∞ , we have the composition $\varphi : M \times \mathbb{R}^n \rightarrow TM$ is C^∞ . The previous claim finishes this result.

Claim: $\Delta M \leq M \times M$ is a regular submanifold.

Proof: Suppose ψ is a chart of $M \times M$. Then so is $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \psi$. Let (U, ϕ) be an atlas of M . Take $\psi = \phi \times \phi : U \times U \rightarrow \mathbb{R}^{2n}$. Then $\Delta M \cap (U \times U) = \{(x, y) \in U \times U : \psi^2(x, y) = 0\}$, since $\psi^2(x, y) = \phi(x) - \phi(y)$, and ϕ is injective.

Claim: The sum function $f : E \oplus E \rightarrow E$ is C^∞ .

Proof: Let U be a trivially open set of E . f restricts to $U \times \mathbb{R}^r \times \mathbb{R}^r \rightarrow U \times \mathbb{R}^r$, where the first component is fixed and $\mathbb{R}^r \times \mathbb{R}^r \rightarrow \mathbb{R}^r$ is addition.

Conversely, suppose $TM \cong M \times \mathbb{R}^r$. Let $\bar{e}_i : M \rightarrow M \times \mathbb{R}^r$ by $\bar{e}_i(p) = (p, e_i)$. $(\bar{e}_i)_{i \in [r]}$ is a global frame of $M \times \mathbb{R}^r \rightarrow M$, so $(s_i)_{i \in [n]}$ is a global frame of $TM \rightarrow M$, where s_i is from the vector bundle isomorphism $M \times \mathbb{R}^r \rightarrow TM$ applied to $\bar{e}_i$.